

My interest in mathematics blossomed in high school as a member of the Stuyvesant High School and NYC math teams. I enjoyed working through difficult problems, understanding unique mathematical structures through the various problem solving techniques that bring them together; I think being able to spend time on hard problems without giving up is one of the most important skills that I developed during my time doing math contests. The process of hard thinking also lends itself to collaboration, as I often stayed after school working on problems with friends, whom I still regularly talk about math with to this day.

Another formative mathematical experience I had in high school was teaching some of the early morning math team classes during my senior year. Each topic has its own interesting scenic overlooks, and I love being a tour guide for my students, pointing them towards nice theorems and important techniques. A favorite of mine was a sequence of lectures I gave to the junior math team on integer partitions, where I discussed various techniques for counting integer partitions and gave two different proofs of Euler's pentagonal theorem over the span of two weeks. I remember spending hours reading through a paper that proved a certain generating function identity and figuring out how to explain the ideas of the proof so that the students could come away with not only a proof, but an intuition for why the identity works, and that thinking was very instructive, improving my understanding of the subject as a whole.

In this vein, an aspect of math that really captivates me is the idea that viewing a concrete phenomenon from the lens of an abstract concept can provide intuition about the true nature of the phenomenon. This kind of thinking especially flourishes in number theory. In high school, I learned about how the Gaussian integers can be used to answer questions like the question of which primes can be represented as a sum of two squares, and I found it incredible that the additional structure of the complex numbers gives so much insight into how sums of two squares behave. Last semester, while sitting in on a number theory lecture, I saw a variation on this theme, where passing to quaternions helps prove that every integer is expressible as a sum of four squares. The past projects enumerated on the website are rife with examples of this too, like the Frobenius problem being generalized to commutative unital rings; I am very intrigued by number theoretic problems like these, and I would be very interested in working on such a project.

This semester, I am participating in an independent reading course with two friends on graph theory and additive combinatorics, roughly following Yufei Zhao's book on the subject. One of the appeals of the subject is the endlessly clever techniques with which we can estimate quantities, for example by analyzing random subgraphs or observing how a vertex with maximal degree relates with the underlying structure. I would love to attend Auburn's REU and work on some graph theory projects, such as questions about edge coloring. One of my classmates briefly discussed anti-Ramsey numbers a few weeks ago, proving an asymptotic bound on the number of colors that could appear in an edge coloring of K_n with no rainbow copy of H as a subgraph, and I would be very excited to explore these ideas more.

I feel that attending Auburn's REU would be beneficial to my future as a mathematician. I plan on attending graduate school and later pursuing a career in academia, and participating in

this program would be an opportunity to see a sliver of math somewhere near the edge of human knowledge, which is an exhilarating prospect to me.